

Set up Earth Engine before Day 1

About 20 minutes. Please do this **at least three days before the course starts** — not the night before, and definitely not on the morning of Day 1.

WHY THE DEADLINE MATTERS

Registration is normally instant, but a small number of accounts hit a snag — usually a university Google account whose administrator blocks the API. If that happens to you on Day 1 you will lose the whole first session. If it happens three days early, we fix it by email in five minutes.

1 What you need

- A **personal Gmail account** — see the warning below.
- A web browser, for Earth Engine and Colab.
- A laptop where you can **install one small application** — Antigravity, your AI assistant for this course. See Part 3 below.
- No credit card. Earth Engine and Antigravity's free tier are both free for this course.

USE A PERSONAL GMAIL, NOT YOUR UNIVERSITY ACCOUNT

University and workplace Google accounts are often configured by an administrator to block third-party API access. You will get as far as the sign-in screen and then hit `Caller does not have required permission`, with no way to fix it yourself.

A personal `@gmail.com` address avoids this entirely. If you do not have one, create one first — it takes two minutes.

2 Register, step by step

1 Sign in to Google with your personal Gmail

If you are already signed in with a university account, either sign out or open a private browsing window. Getting this wrong here causes most of the problems later.

2 Go to the Earth Engine page in Google Cloud

console.cloud.google.com/earth-engine

You may be asked to accept the Google Cloud terms of service. Accept them. This does not cost anything and does not require payment details.

3 Create a Cloud project

Earth Engine organises access around a "project". Create one and give it a name you will recognise — for example `ee-yourname`.

Google will generate a **project ID** from that name, sometimes adding numbers to make it unique. **The ID is what you need later, not the display name.** They are often different.

4 Register the project for noncommercial use

Choose **Noncommercial / academic** when asked what the project is for, and answer the short eligibility questionnaire. You are a student using this for coursework, which qualifies.

You will be placed on the **Community Tier**, which gives you 150 EECU-hours of computing per month at no cost. That is far more than this course needs.

5 Write your project ID down

You will type it into every notebook, in every session, for the whole course. Put it somewhere you can copy from — a note on your phone is fine.

MY EARTH ENGINE PROJECT ID

6 Run the check notebook

Open `00_check_gee_access.ipynb` in Google Colab, put your project ID in the one place it asks, and run the single cell. This is the only proof that setup worked.

You are done when it prints **ALL CHECKS PASSED**.

3 Install Antigravity — your AI assistant

This course uses **Google Antigravity** as the free AI assistant you will prompt to write Earth Engine code. Unlike Claude or ChatGPT, it is a small application you install on your laptop rather than a website — but it costs nothing and needs no credit card.

1 Download and install

Go to antigravity.google and download the version for your operating system:

macOS	12 (Monterey) or newer
Windows	Windows 10 or newer, 64-bit
Linux	64-bit; <code>.deb</code> for Ubuntu/Debian, <code>.rpm</code> for Fedora/Red Hat

You also need Google Chrome installed — Antigravity uses it for browser-integration features.

2 Sign in with the same personal Gmail

Open Antigravity and sign in with the **same personal Gmail** you used for Earth Engine. One account, no separate password to remember.

3 Start a chat — do not run notebooks inside Antigravity

Create a new project (any empty folder), then look for the chat panel and start typing — this is where you will paste your PROMPT-recipe prompts.

IMPORTANT — THIS IS NOT HOW YOU WILL USE IT

Antigravity can also open, edit and run files on its own. **Do not use that for this course.** Use only its chat: ask for code, read the answer, then **copy the code into your Colab notebook and run it there**, exactly as you would with Claude or Gemini. Every lab in this course is built and tested in Colab — that is where your maps will actually display.

4 If the check fails

Find your message below. These cover essentially everything that goes wrong.

WHAT YOU SEE	WHAT TO DO
Project 'projects/xxx' not found or deleted	Your project ID is wrong. Go back to the Cloud console and copy the ID , not the display name. It often has extra digits on the end.
Caller does not have required permission or the sign-in page refuses you	You are almost certainly on a university account. Switch to a personal Gmail and register again from step 1.
Earth Engine API has not been used in project ... before or it is disabled	The API is not switched on. Open the link in the error message and click Enable , then wait a minute and re-run.
not signed up for Earth Engine	The project exists but was never registered. Return to step 4.
The authentication link does nothing	Pop-ups are blocked, or you are signed into several Google accounts at once. Use a private browsing window with only the personal Gmail signed in.
Something else entirely	Email your instructor the whole error message — a screenshot of the full red text, not just the last line — plus your project ID. Do this before Day 1.
Antigravity install is blocked by your laptop	Some managed/work laptops block new installs. Use a personal laptop if you have one, or ask your IT admin for permission ahead of time — not on the morning of Day 1.
Antigravity asks for a work/organisation account	Sign out and sign back in with your personal Gmail — the same one you used for Earth Engine.

5 Two things worth knowing before Day 1

You will re-authenticate every session

Colab gives you a temporary machine that gets recycled when you are idle. Every notebook in this course therefore starts with the same two lines, and you will run them many times:

```
ee.Authenticate()  
ee.Initialize(project='ee-yourname')
```

This is normal and not a sign anything is broken.

Computing is not free, even when you pay nothing

Your 150 EECU-hours per month are a real budget. Two habits keep you comfortably inside it:

- Study **one province**, not a whole country.
- Always pass an explicit `scale=` — `100` metres is right for area statistics.

We will explain both properly on Day 1. For now, just know that "make it bigger and more detailed" has a cost attached.

6 What to bring

- A laptop with a charger. A tablet will not work for this.
- Your project ID, written down.
- **Antigravity installed and signed in** — see Part 3. This is your AI assistant for the course. Gemini is also built into Colab as a backup, and needs no separate account.
- **The name of a province you actually care about.** Every lab lets you swap in your own study area, and the course is considerably more interesting when the map is somewhere that means something to you.

YOU DO NOT NEED TO KNOW HOW TO CODE

This course assumes no programming background and very little GIS. You will describe what you want in English, an AI will write the code, and you will learn to tell whether the answer is right. That last part is the actual skill, and it does not require you to write Python.